Chapter

Sulphuric Acid

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1. Superphosphate of lime is a : SQP 2023

(a) Fertiliser
(b) Deterg ent
(c) Explosive
(d) Battery

2. Chemical name of oleum is: MAIN 2021

(a) Hydrated sulphuric acid
(b) Pyrosulphuric acid
(b) Acetyl aldehyde
(d) Ortho sulphuric

3. Gas obtained by dehydration of Ethanol COMP 2021

4. A black coloured substance formed when Sugar is dehydrated. MAIN 2023

5. $K_2CO_3 + H_2SO_4 \longrightarrow$

6. $Na_2SO_3 + H_2SO_4 \longrightarrow .... + ... + ...$

7. $Zn + \underset{\text{(Conc.)}}{H_2SO_4} \longrightarrow$

8. $CuSO_4 \cdot 5H_2O \xrightarrow[\Delta]{\text{Conc.} H_2SO_4}$

9. In this question you are required to supply the word for words that will make each sentence into correct statement which is to be written down again completely.
(i) Copper sulphate crystals are dehydrated by sulphuric acid.
(ii) Crystals of sulphur are obtained when a solution of sulphur in carbon disulphide is allowed to evaporate. MAIN 2023

10. Give one equation each to show the following properties of sulphuric acid:
(i) Dehydrating property.
(ii) Acidic nature.
(iii) As a non-volatile acid. MAIN 2014

11. Give examples of the use of sulphuric acid. SQP 2022

(a) As an electrolyte in everyday use.
(b) As non-volatile acid.
(c) As an oxidising agent.

12. A colourless oily liquid was poured slowly into a large volume of cold water. With constant stirring the temperature of the water rose considerably. On testing the solution, only hydrogen and sulphate ions could be detected.

(i) How would you carry out the test for the presence of hydrogen ions ?
(ii) Name the oily liquid. COMP 2022

13. Name the process used for the large scale manufacture of Sulphuric acid.
MAIN 2023

14. When hot concentrated sulphuric acid is added to sulphur, it gets oxidised to ______ (H_2S/SO_2) SQP 2022

Solution

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1. (a) Fertiliser

Superphosphate of lime is a mixture of **calcium dihydrogen phosphate ($Ca(H_2PO_4)_2$)** and **calcium sulfate ($CaSO_4$, gypsum)**. It is produced by the action of **sulphuric acid on phosphate rock** and is an important **fertiliser** that provides **phosphorus** for plant growth.

2. (b) Pyrosulphuric acid

Explanation:
Oleum is chemically known as **pyrosulphuric acid ($H_2S_2O_7$)**. It is formed when **sulfur trioxide (SO_3)** reacts with **concentrated sulfuric acid (H_2SO_4)**:

$$H_2SO_4 + SO_3 \longrightarrow H_2S_2O_7$$

3. Ethene or Ethylene

4. Carbon/Sugar Charcoal

5. $Na_2SO_3 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O + SO_2\uparrow$

6. $Na_2SO_3H_2SO_4 \longrightarrow Na_2SO_4 + H_2O + SO_2$

7. $Zn + \underset{\text{(Conc.)}}{2H_2SO_4} \longrightarrow ZnSO_4 + SO_2\uparrow + 2H_2O$

8. $CuSO_4 \cdot 5H_2O \xrightarrow[\Delta]{\text{Conc.}H_2SO_4} CuSO_4 + 5H_2O$

9. (i) Copper sulphate crystals are dehydrated by **concentrated sulphuric acid.**

(ii) Crystals of **rhombic sulphur** are obtained when a solution of sulphur in carbon disulphide is allowed to evaporate.

10.

(i) $C_{12}H_{22}O_{11} + H_2SO_4 \longrightarrow 12C + [11H_2O.H_2SO_4]$

(ii) $CuO + H_2SO_4 \longrightarrow CuSO_4 + H_2O$

(iii) $NaCl + H_2SO_4 \xrightarrow[200°C]{Below} NaHSO_4 + HCl$

11. (a) As an electrolyte: Sulphuric acid is used as the **electrolyte in storage batteries.**

(b) As a non-volatile acid: Sulphuric acid is used in the **preparation of hydrochloric acid, nitric acid, and acetic acid.**

(c) As an oxidising agent: Sulphuric acid is used in the **preparation of sulphur dioxide (SO_2).**

12. (i) Test for hydrogen ions: Take a small portion of the solution in a test tube and add a pinch of **sodium carbonate (Na_2CO_3)**. If **hydrogen ions (H^+)** are present, **carbon dioxide (CO_2) gas** is evolved with brisk effervescence:

$$2H^+ + Na_2CO_3 \longrightarrow 2Na^+ + H_2O + CO_2 \uparrow$$

Pass the evolved CO_2 through **lime water ($Ca(OH)_2$)**, which turns **milky**, confirming the presence of H^+ ions:

$$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 \downarrow + H_2O$$

(ii) Name of the oily liquid: The liquid is **concentrated sulphuric acid (H_2SO_4)**, which shows **exothermic dissolution in water** and provides **H^+ and SO_4^{2-} ions** in solution.

13. Contact process

14. SO_2